Course Syllabus

Bioinformatics Engineering

ENBC455
Spring 2026

Learning Outcomes

Bioinformatics is an extremely broad field that seeks to "make sense of the enormous quantities of sequence data and structural data generated" by using "computer programs that help to reveal the fundamental mechanisms underlying biological problems." As a biocomputational engineer, you are uniquely suited to bridge the divide in bioinformatics between biologists/geneticists and programmers/data scientists and uniquely positioned to quickly advance in this area of study. This introductory course covers the core principles of bioinformatics while encouraging students to apply their programming skills to real-world biological problems.

After successfully completing this course, you will be able to:

- Use publicly available resources to study nucleic acids and proteins and analyze their sequences.
- Process high-throughput sequencing data
- Solve bioinformatic problems with Python.
- Teach yourself how to utilize new bioinformatic tools.

Required Resources

OPTIONAL Textbook: The Biostar Handbook. 2nd Edition. (Purchase here for six months of online access and to download the PDF to keep forever.)

Course website: **elms.umd.edu**

If you are having difficulty with ELMS/Canvas, please login to ELMS/Canvas to access 24/7 Canvas Chat Support or call: (877) 399-4090.

Technical Assistance: If you experience problems with your computer, contact the Division of Information Technology service desk. Location: McKeldin Library, Room 1221. E-mail: itsupport@umd.edu. Website and live chat: https://itsupport.umd.edu/itsupport/en. Phone: (301) 405-1500.

Dr. Jarred Callura
jcallura@umd.edu

Class Meets
Tuesday, Thursday
10:00-11:15 a.m.
Room: IV-4321

Office Hours
By appointment

Prerequisites
N/A

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Completion of ENBC311 with a grade of "C-" or better; or permission of Fischell Department of Bioengineering department. Must be in the Biocomputational Engineering program.

Course Communication
Time-sensitive information will be sent to students via an ELMS announcement. Please contact the instructor via email to discuss questions or accommodations. Below is a link with helpful guidance on writing professional emails (ter.ps/email).

Campus Policies

It is the students' and my responsibility to know and abide by the University of Maryland's policies that relate to all courses, which include topics like:

- Academic integrity
- Student and instructor conduct
- Accessibility and accommodations
- Attendance and excused absences
- Grades and appeals
- Copyright and intellectual property

Please visit www.ugst.umd.edu/courserelatedpolicies.html for the Office of Undergraduate Studies' full list of campus-wide policies and follow up with me if you have questions.

Course-Specific Policies and Responsibilities

Academic Integrity: I, like the University, take academic dishonesty very seriously, and have a zero-tolerance policy. That means, regardless of the weight of the assignment, if it is suspected that the work you turn-in is not your own, I may submit it to the Student Honor Council for evaluation without warning. For more information regarding the University's academic integrity policy, please consult the Student Honor Council: https://studentconduct.umd.edu/.

Learner Responsibilities: As a student in this class, I expect you to:

- take responsibility for your own learning
- be prepared and enthusiastically participate
- treat others with tolerance and respect
- set high standards for your work

Instructor Responsibilities: I commit to communicating openly and frequently with you about this class. I will maintain a professional, safe learning environment adhering to the policies of the University. You can expect a reply to communication, be it via e-mail or online discussions, within 24 hours. I will make every attempt to respond more quickly, but please do not procrastinate if you plan on asking for help or on scheduling office hours.

Communication with Peers

With a diversity of perspectives and experience, we may find ourselves in disagreement and/or debate with one another. As such, it is important that we agree to conduct ourselves in a professional manner and that we work together to foster and preserve a virtual classroom environment in which we can respectfully discuss and deliberate controversial questions.

With a respect for reasoned argument, I encourage you to confidently exercise your right to free speech—bearing in mind, of course, that you will be expected to craft and defend arguments that support your position. Keep in mind that free speech has its limit, and this course is NOT the space for hate speech, harassment, and derogatory language. I will make every reasonable attempt to create an atmosphere in which each student feels comfortable voicing their argument without fear of being personally attacked, mocked, demeaned, or devalued.

Any behavior (including harassment, sexual harassment, and racially and/or culturally derogatory language) that threatens this atmosphere will not be tolerated. Please alert me immediately if you feel threatened, dismissed, or silenced at any point during our semester together and/or if your engagement in discussion has been in some way hindered by the learning environment.

Communication with Instructor

Email: If you need to reach out and communicate with me, please email me at **jcallura@umd.edu**. Please reach out about personal, academic, and intellectual concerns/questions. Again, I will try my best to, and usually do, respond to emails quickly.

ELMS: I will send important announcements via ELMS messaging. Please make sure that your email and announcement notifications (including changes in assignments and/or due dates) are enabled in ELMS so that you do not miss any messages.

Course Structure

Lectures: There will be two 75-minute lecture periods per week. The lecture time will consist of reviewing bioinformatics concepts and exploring bioinformatics tools. Students are expected to read the material to be covered that day before class and bring their computer to class. Computers are mandatory for lecture participation.

Class Participation: Students will be expected to participate in class and during group problem solving activities. Each lecture will have several questions that require students to respond via the PointSolutions app or web software. Students should bring a web-enabled device (smartphone, iPad, laptop) to use PointSolutions. Responses will be counted for completion, not for correctness. You must attend lecture in person to receive credit. Students may miss a lecture without penalty due to an excused absence, assuming proper documentation is provided. Everyone receives one penalty-free absence.

Homework: In weekly homework assignments, you will apply your Python programming skills to bioinformatics. All homework assignments are available at the start of the course and can be completed well before their due date. Although the lectures do not include Python information, please do not hesitate to contact me for homework help.

Gradescope: You will submit the homework assignments via Gradescope. Please let me know if you have any upload issues or questions about the platform.

Exams: There will be three closed-note exams. These exams will assess your understanding of the lecture material, and in-class activities will prepare you for the exams. Exams are broken into two parts. Part I is multiple-choice and short answer questions to be completed without a computer, and Part II is questions requiring a computer asking you to perform some bioinformatics analyses. The Final Exam will not be cumulative. Note: students who are scheduled for more than three exams in one day can request an accommodation.

Projects: Four projects will be assigned throughout the semester. The Study-A-Gene project will have you perform research on a human gene of interest. The Find-A-Gene project will have you search through real biological data for an uncharacterized gene, identify such a gene, and then analyze it using the various tools we will cover in the course. The Assemble-A-Genome project will have you determine the genome sequence of an organism using raw DNA sequencing data. The Quantify-A-Transcript project will have you perform differential expression analysis on real RNA sequencing data. The Find-A-Gene project will be team-based and includes a presentation component.

Late policy: I want you to be accountable for assignment due dates, therefore I have a NON-NEGOTIABLE late-work policy: Assignments submitted late without prior arrangement for an extension not be accepted. If you do not believe you can complete an assignment on time due to extenuating circumstances, you must communicate that to me BEFORE the due date, and I will provide an appropriate extension.

A.I. policy: In this course, I encourage you to use artificial intelligence-powered programs, such as ChatGPT, to help you with some assignments. When you use these tools, it is your responsibility as a scholar to make sure you are clearly communicating the A.I. involvement in your work. Please review the instructions in each assignment for more details on whether using A.I. is allowed and how to show your work.

Grades

Grades are not given but earned. Your grade is determined by your performance on the learning assessments in the course and is assigned individually. If earning a particular grade is important to you, please speak with me at the beginning of the semester to receive some helpful suggestions for achieving your goal.

All assessment scores will be posted on the course ELMS page. If you would like to review any of your grades (including the exams), or have questions about how something was scored, please email me to schedule a time for discussion. Late work will not be accepted for course credit, so please plan to have it submitted well before the scheduled deadline. I am happy to discuss any of your grades with you, and if I have made a mistake, then I will immediately correct it. Any formal grade disputes must be submitted in writing and within one week of receiving the grade.

Learning Assessments	Category Weight
Lecture Participation	5%
Homework	15%
Exams	30%
Projects	50%

Final letter grades are assigned based on the percentage of total assessment points earned. I may, at my discretion, lower the below grade boundaries during the semester, thus making grading more lenient. A +/– grading scheme for final grades will be used.

Final Grade Cutoffs									
+	97.00%	+	87.00%	+	77.00%	+	67.00%		
A	94.00%	B	84.00%	C	74.00%	D	64.00%	F	<60.0%
-	90.00%	-	80.00%	-	70.00%	-	60.00%		

Course Schedule

Week	Class Topic
1	**Course introduction. Databases.**
2	**Accessing databases. Genome browsers.**
3	**Pairwise sequence alignment. Scoring matrices.**
4	**Alignment algorithms. Statistical analyses of alignments.**
5	**Midterm Exam 1** **BLAST; Multiple sequence alignment.**
6	**Multiple sequence alignment algorithms. Benchmarking and molecular evolution.**
7	**Phylogenetic analyses. Tree construction.**
8	**Spring Break**
9	**Midterm Exam 2** **Tree evaluation. High-throughput sequencing.**
10	**NGS technologies. Find-A-Gene project presentations.**
11	**NGS data analysis. Genome assembly.**
12	**Mapping. SAM and BAM files.**
13	**Variant calling. Variant effect prediction.**
14	**RNA sequencing (RNA-seq). Quantifying transcript abundance.**
15	**Differential expression analysis. Interpreting RNA-seq results.**
16	**Final Exam**

Note: I retain the right to make changes to the course calendar based on the timeline of the class, feedback from students, and/or logistical issues. Please monitor the course ELMS page for current deadlines.

Resources & Accommodations

Accessibility & Disability Services

The University of Maryland is committed to creating and maintaining a welcoming and inclusive educational, working, and living environment for people of all abilities. The University of Maryland is also committed to the principle that no qualified individual with a disability shall, on the basis of disability, be excluded from participation in or be denied the benefits of the services, programs, or activities of the University, or be subjected to discrimination. The **Accessibility & Disability Service (ADS)** provides reasonable accommodations to qualified individuals to provide equal access to services, programs, and activities. ADS cannot assist retroactively, so it is generally best to request accommodations several weeks before the semester begins or as soon as a disability becomes known. Any student who needs accommodations should contact me as soon as possible so that I have sufficient time to make the requested arrangements.

For assistance in obtaining an accommodation, contact Accessibility and Disability Service at 301-314-7682, or email them at **adsfrontdesk@umd.edu**.

Student Resources and Services

Taking personal responsibility for your own learning means acknowledging when your performance does not match your goals and doing something about it. I hope you will come talk to me so that I can help you find the right approach to success in this course, and I encourage you to visit tutoring.umd.edu to learn more about the wide range of campus resources available to you. In particular, everyone can use some help sharpen their communication skills (and improving their grade) by visiting https://english.umd.edu/writing-programs/writing-center and schedule an appointment with the campus Writing Center. You should also know there are a wide range of resources to support you with whatever you might need (see go.umd.edu/assistance), and if you just need someone to talk to, visit counseling.umd.edu or one of the many other resources on campus. Most services free because you have already paid for it, and **everyone needs help**… all you must do is ask for it.

Basic Needs Security

If you have difficulty affording groceries or accessing sufficient food to eat every day, or lack a safe and stable place to live and believe this may affect your performance in this course, please visit https://studentaffairs.umd.edu/support-resources/thrive-center-for-essential-needs for information about resources the campus offers you and let me know if I can help in any way.